

Cryptanalysis and GPU-Accelerated Remediation of a Chaotic DNA Image Cryptosystem: Resolving Theoretical Information Loss and Diffusion Collapse in Yousif et al.

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Abstract

The rapid expansion of high-throughput multimedia transmission across telemedicine, autonomous systems, and secure teleconferencing demands image cryptosystems capable of destroying strong spatial correlation without imposing latency bottlenecks. Cryptosystems combining continuous chaotic dynamical systems with biological DNA computing have attracted extensive research due to their infinite continuous phase space and intrinsic parallel confusion. In this paper, we present a comprehensive cryptanalysis and high-performance CUDA implementation of the prominent peer-reviewed image cryptosystem proposed by Yousif et al. (*Multimedia Tools and Applications*, Springer, 2022).

Our mathematical investigation exposes two fatal theoretical contradictions in the published literature:

1. **The 50% Information Loss Flaw:** The paper splits one plaintext image into two separate images, but then combines them using DNA addition. Adding two DNA bases causes irreversible data loss because multiple distinct base pairs yield the exact same sum. Consequently, the receiver cannot reconstruct the original image from a single ciphertext without ambiguous guessing—a fatal breakdown the authors quietly concealed by feeding a second cipher image into the receiver in their own flowchart (Figure 7).
2. **The Plaintext Avalanche Diffusion Collapse:** The paper claims near-ideal plaintext sensitivity ($\text{NPCR} > 99.61\%$). However, its diffusion step operates strictly along isolated rows using bitwise XOR without cross-row coupling or arithmetic carries. A single-pixel change can alter at most a single row ($\leq 0.0926\%$ of pixels in Full HD 1920×1080). We show that the authors mistakenly evaluated key sensitivity and mislabeled it as plaintext sensitivity.

To remediate these vulnerabilities, we design and implement an end-to-end hardware-software co-design: an enhanced 4-pass bi-directional 2D Addition-Rotation-XOR (ARX) diffusion

engine running natively on NVIDIA GPUs. Our system achieves bit-exact lossless recovery ($\text{MSE} = 0.0000$), near-ideal differential security ($\text{NPCR} = 99.608\% - 99.626\%$, $\text{UACI} = 33.442\% - 33.478\%$), near-optimal entropy ($H > 7.9999$), zero cross-pixel correlation ($|r_{xy}| < 0.02$), and real-time execution speeds on Full HD 1920×1080 video frames.

Keywords—Image Encryption, Cryptanalysis, Chaotic Dynamics, DNA Computing, CUDA GPU Acceleration, ARX Primitives, Differential Avalanche, Ciphertext Expansion.

1 Introduction and Research Motivation

Digital multimedia streams represent the dominant share of global internet bandwidth. High-definition (1080p) and 4K digital video feeds generate hundreds of megabytes per second, creating acute transmission bottlenecks. Uncompressed image and video data possess distinct topological features that separate them from alphanumeric text:

- **High Spatial Redundancy:** Adjacent pixels in natural images exhibit exceptionally strong cross-correlation across horizontal, vertical, and diagonal directions, with Pearson correlation coefficients routinely exceeding $r_{xy} > 0.98$.
- **Bulk Data Volumes:** A standard Full HD 1920×1080 8-bit frame contains over two million bytes (2.07×10^6 bytes, 1.66×10^7 bits). In RGB color video, each frame represents over 6.22 MB. Encrypting 60 frames per second requires sustaining cipher throughput exceeding 370 MB/s with latency bounded strictly below 16.6 ms per frame.
- **Statistical Vulnerability:** Applying standard symmetric block ciphers like AES in Electronic Codebook (ECB) mode preserves global contours and textures because identical plaintext blocks encrypt to identical ciphertext blocks.

While chained modes such as Cipher Block Chaining (CBC) or Galois/Counter Mode (GCM) prevent pattern leakage, their sequential dependencies restrict parallel GPU thread dispatch, causing significant CPU bottlenecks in multi-camera or real-time streaming architectures.

1.1 The Chaotic DNA Cryptographic Paradigm

To achieve both high cryptographic confusion and massive parallel performance, research has focused on combining **Continuous Chaotic Dynamical Systems** with **Biological DNA Computing**.

Continuous chaotic attractors (such as the 3D Chen system and the 4D Lorenz hyperchaotic system) are governed by coupled non-linear differential equations. They exhibit ergodicity, non-periodicity, and extreme sensitivity to initial conditions (the *Butterfly Effect*). A secret key difference of $\Delta x_0 = 10^{-16}$ produces exponential trajectory divergence ($e^{\lambda t}$), spanning an effectively infinite continuous phase space ($> 2^{256}$) impervious to brute-force search.

Simultaneously, DNA computing abstracts digital data into the quaternary alphabet of nucleic acid bases: Adenine (A), Cytosine (C), Guanine (G), and Thymine (T). Operating over Galois Field $\text{GF}(4)$ and modular ring $\mathbb{Z}_4$, DNA biological addition and dynamic rule assignment introduce non-linear confusion that destroys byte-level frequency signatures and prevents linear cryptanalysis.

1.2 The Baseline Literature Under Investigation

In 2022, Yousif et al. published a prominent image encryption algorithm in Springer’s *Multimedia Tools and Applications*, titled “A new image encryption based on bit replacing, chaos and DNA coding techniques” [1]. The authors proposed an architecture based on four sequential components:

1. **Bit Replacing Technique (BRT):** Splitting the 8-bit image into two separate 8-bit images (U_1 and U_2) to segregate the four Most Significant Bits (MSBs) from the four Least Significant Bits (LSBs).
2. **Dual Permutation Pipeline:** Coordinate scrambling using the 2D Arnold Cat Map followed by 3D Chen chaotic sequence sorting.
3. **Quaternary DNA Confusion:** Dynamic encoding of both images into DNA matrices, followed by biological DNA addition modulo 4.
4. **1D Keystream Diffusion:** Sequential column-wise bitwise XOR diffusion driven by a Lorenz hyperchaotic keystream (Equation 9 in [1]).

In Table 6 of their paper, the authors reported stellar differential benchmarks: NPCR = 99.6105% – 99.6215%, UACI = 33.44% – 33.70%, and claimed perfect lossless decryption from a single transmitted ciphertext image.

1.3 The Reproduction Dilemma & Our Contributions

Our initial goal was to construct a massively parallel CUDA C++ implementation of Yousif et al.’s cryptosystem to enable real-time video stream protection on modern NVIDIA GPUs.

However, when we strictly implemented the algorithm according to the published mathematical equations, our automated testing suite uncovered two fundamental theoretical contradictions:

1. **Mathematical Impossibility of Decryption from One Ciphertext:** Step 13 of the paper adds two DNA matrices modulo 4. We prove that this non-injectively maps 16 input base pairs down to only 4 sums, causing a permanent loss of 2^{8MN} distinct states by the Pigeonhole Principle. Decrypting from a single ciphertext is mathematically impossible. We show that the authors concealed this in their text while quietly feeding a second cipher image into the receiver’s flowchart (Figure 7 in [1]).

2. **Mathematical Impossibility of Avalanche Diffusion:** The paper’s 1D XOR recurrence (Eq. 9) is strictly row-decoupled: a perturbed pixel can never propagate beyond its single row. The maximum theoretical NPCR is bounded by $1/M \approx 0.0976\%$ (compared to the claimed 99.61%), and bitwise XOR lacks arithmetic carries, bounding UACI to 0.00038%. We show the authors ran a Key Sensitivity test and mislabeled it as Plaintext Sensitivity.

To resolve these breakdowns, we engineered a novel 4-pass bi-directional 2D Addition-Rotation-XOR (ARX) diffusion engine natively on NVIDIA CUDA. This paper documents the complete mathematical proofs of the paper’s flaws, our 2D ARX resolution, and the full experimental validation verifying bit-exact lossless recovery ($\text{MSE} = 0.0000$) and real-time throughput (> 200 FPS).

2 Theoretical Foundations: Chaos and DNA Computing

2.1 Continuous Chaotic Dynamical Systems

2.1.1 The 3D Chen Chaotic Attractor

Discovered by Chen and Ueta in 1999 [3], the 3D Chen system is defined in autonomous phase space by the non-linear ODE system:

$$\begin{cases} \frac{dx}{dt} = a(y - x) \\ \frac{dy}{dt} = (c - a)x - xz + cy \\ \frac{dz}{dt} = xy - bz \end{cases} \quad (1)$$

Under parameter values $a = 35$, $b = 3$, and $c = 28$, the system exhibits a large maximal positive Lyapunov exponent $\lambda_1 = 2.027$. This ensures rapid spatial divergence of trajectories.

In our CUDA engine, the Chen system is integrated numerically using Fourth-Order Runge-Kutta (RK4) with fixed step size $h = 0.001$:

$$\mathbf{X}_{n+1} = \mathbf{X}_n + \frac{h}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \quad (2)$$

where $\mathbf{X} = [x, y, z]^T$. Sorting the continuous trajectory $x(t)$ in ascending order generates coordinate permutation lookup tables that shuffle pixel coordinates on the GPU.

2.1.2 The 4D Lorenz Hyperchaotic Attractor

To generate keystreams for dynamic DNA rule selection and non-linear pixel diffusion, we deploy the 4D Lorenz hyperchaotic system [4]:

$$\begin{cases} \frac{dx}{dt} = a(y - x) + w \\ \frac{dy}{dt} = cx - y - xz \\ \frac{dz}{dt} = xy - bz \\ \frac{dw}{dt} = -dw \end{cases} \quad (3)$$

When initialized with parameters $a = 10$, $b = 8/3$, $c = 28$, and $d = 1$, the system exhibits **two positive Lyapunov exponents**:

$$\lambda_1 = 0.54, \quad \lambda_2 = 0.17, \quad \lambda_3 = 0, \quad \lambda_4 = -11.71 \quad (4)$$

Because hyperchaotic trajectories expand simultaneously along two orthogonal directions in phase space, they generate four coupled pseudo-random streams (x, y, z, w) that resist phase-space attractor reconstruction attacks.

2.2 DNA Computing and Biological Quaternary Encoding

DNA computing represents information using the quaternary alphabet of nucleic acid bases: Adenine (A), Cytosine (C), Guanine (G), and Thymine (T) [5].

Definition 1 (DNA Quaternary Encoding) *Each nucleic acid base encodes exactly two binary bits:*

$$\mathcal{B} = \{A, C, G, T\} \longleftrightarrow \{00_2, 01_2, 10_2, 11_2\} \quad (5)$$

An 8-bit grayscale pixel $P(i, j) \in [0, 255]$ is encoded as four consecutive DNA bases:

$$P(i, j) = ((b_7b_6)_2, (b_5b_4)_2, (b_3b_2)_2, (b_1b_0)_2) \in \mathcal{B}^4 \quad (6)$$

Theorem 1 (Watson-Crick Complementarity) *Under the biological Watson-Crick double helix model [6], Adenine pairs strictly with Thymine ($A \leftrightarrow T$), and Cytosine pairs strictly with Guanine ($C \leftrightarrow G$). Let $v(B) \in \{0, 1, 2, 3\}$ denote the 2-bit integer value of base B . An encoding rule is biologically valid if and only if:*

$$v(B) \oplus v(\overline{B}) = 11_2 = 3 \quad (7)$$

Out of $4! = 24$ conceivable bijective mappings, exactly **eight** satisfy this biological constraint.

The four 2-bit integers $\{0, 1, 2, 3\}$ form two complementary pairs whose bitwise XOR equals 3:

$$\mathcal{P}_1 = \{0, 3\} = \{00_2, 11_2\}, \quad \mathcal{P}_2 = \{1, 2\} = \{01_2, 10_2\} \quad (8)$$

To satisfy Equation (7):

1. The base pair $\{A, T\}$ must be assigned exclusively to either $\mathcal{P}_1$ or $\mathcal{P}_2$, yielding $\binom{2}{1} = 2$ choices.
2. Within the chosen pair, there are $2! = 2$ ways to assign values to A and T (either $v(A) = 0, v(T) = 3$ or $v(A) = 3, v(T) = 0$).
3. The remaining pair is assigned to $\{C, G\}$, which can also be ordered in $2! = 2$ ways.

Multiplying these independent choices gives $N_{\text{rules}} = 2 \times 2 \times 2 = 8$. The remaining 16 permutations violate chemical base pairing.

The eight valid biological rules are compiled in Table 1.

Table 1: The Eight Valid Biological DNA Encoding Rules

Rule Index	00_2	01_2	10_2	11_2
Rule 1	A	C	G	T
Rule 2	A	G	C	T
Rule 3	C	A	T	G
Rule 4	C	T	A	G
Rule 5	G	A	T	C
Rule 6	G	T	A	C
Rule 7	T	C	G	A
Rule 8	T	G	C	A

2.3 DNA Algebraic Operations in $\text{GF}(4)$

DNA addition $\boxplus$ and DNA subtraction $\boxminus$ are evaluated over $\mathbb{Z}_4$ as:

$$B_1 \boxplus B_2 = R^{-1} \left((R(B_1) + R(B_2)) \pmod{4} \right) \quad (9)$$

$$B_1 \boxminus B_2 = R^{-1} \left((R(B_1) - R(B_2) + 4) \pmod{4} \right) \quad (10)$$

The Cayley operation tables for Rule 1 ($A = 0, C = 1, G = 2, T = 3$) are shown in Table 2.

Table 2: Cayley Addition ($\boxplus$) and Subtraction ($\boxminus$) Tables for Rule 1

$\boxplus$	A	C	G	T	$\boxminus$	A	C	G	T
A	A	C	G	T	A	A	T	G	C
C	C	G	T	A	C	C	A	T	G
G	G	T	A	C	G	G	C	A	T
T	T	A	C	G	T	T	G	C	A

3 Deconstruction of the Baseline Algorithm (Yousif et al., 2022)

Let $I \in \mathbb{Z}_{256}^{M \times N}$ denote the 8-bit plaintext image of size $M \times N$, representing $8MN$ bits.

3.1 Algorithmic Pipeline Specification

3.1.1 Step 1: Bit Replacing Technique (BRT)

In Section 3.1 and Section 3.2 of [1], the plaintext image I undergoes a 1-bit to 2-bit expansion mapping known as the Bit Replacing Technique (BRT). Rather than trivial masking with zero padding, each individual bit $b \in \{0, 1\}$ of an 8-bit pixel is replaced by a distinct 2-bit string:

$$\phi(b) = \begin{cases} 10_2 & \text{if } b = 0 \\ 01_2 & \text{if } b = 1 \end{cases} \quad (11)$$

Equivalently, this mapping is expressed algebraically over the integer value as $\phi(b) = 2 - b$.

As illustrated in Section 3.1 of [1], an 8-bit pixel such as $97 = 01100001_2$ expands into 16 binary bits:

$$\begin{aligned} \phi(01100001_2) &= \phi(0) \phi(1) \phi(1) \phi(0) \phi(0) \phi(0) \phi(0) \phi(1) \\ &= \mathbf{10\ 01\ 01\ 10\ 10\ 10\ 10\ 01} \end{aligned} \quad (12)$$

These 16 bits are then split into two distinct 8-bit bytes (in this example, yielding decimal values 150 and 169). Across the image, each pixel's 8 bits are partitioned into two 4-bit nibbles which each expand into an 8-bit byte ($4 \times 2 = 8$ bits), producing two full 8-bit matrices $U_1, U_2 \in \mathbb{Z}_{256}^{M \times N}$:

$$U_1(i, j) = \sum_{k=0}^3 \phi((I(i, j) \gg k) \& 1) \cdot 4^k \quad (13)$$

$$U_2(i, j) = \sum_{k=0}^3 \phi((I(i, j) \gg (k + 4)) \& 1) \cdot 4^k \quad (14)$$

Notice that neither U_1 nor U_2 involves zero padding: every single bit position in the resulting bytes is actively occupied by the expanded chaotic pairs.

Remark 1 *The Bit Replacing Technique expands each 8-bit pixel ($8MN$ bits) into two complete 8-bit bytes ($16MN$ bits), fundamentally doubling the data volume from $M \times N \times 8$ bits to $2 \times M \times N \times 8$ bits prior to chaotic permutation and DNA encoding.*

3.1.2 Step 2: Chen Chaotic Coordinate Permutation (Collision-Free Sorting)

In Steps 4–7 of [1], coordinate permutation is achieved using the continuous 3D Chen chaotic system. The continuous chaotic trajectory generates a floating-point sequence $X_1 = \{x_1, x_2, \dots, x_{MN}\}$. Rather than using a direct modular mapping (which would induce destination collisions whenever different coordinates map to the same modular output), the algorithm extracts a permutation by **sorting the chaotic array**:

$$[L', \pi] = \text{sort}(X_1) \quad (15)$$

where $\pi \in \mathcal{S}_{MN}$ is the permutation vector of original indices, representing a bijection $\pi : \{1, \dots, MN\} \leftrightarrow \{1, \dots, MN\}$.

Lemma 2 (Collision-Free Permutation Guarantee) *Because π is obtained by an argsort permutation of the unique index array, every target position appears exactly once:*

$$\pi(i) = \pi(j) \iff i = j \quad (16)$$

The permutation is mathematically guaranteed to be strictly collision-free with zero coordinate overwrites. In our GPU implementation, this is executed via high-throughput 64-bit Radix Sort on IEEE-754 mantissa state.

Both intermediate matrices U_1 and U_2 are scrambled according to π :

$$E_1 = \pi(U_1), \quad E_2 = \pi(U_2) \quad (17)$$

3.1.3 Step 3: 1D Keystream XOR Diffusion

In Steps 8–10 (Equation 9 of [1]), the 4D Lorenz hyperchaotic system generates a diffusion keystream $Y_3 \in \mathbb{Z}_{256}^{M \times N}$. Sequential 1D column/row diffusion is applied to both scrambled matrices:

$$E'_1(i, j) = E'_1(i, j-1) \oplus E_1(i, j) \oplus Y_3(i, j) \quad (18)$$

$$E'_2(i, j) = E'_2(i, j-1) \oplus E_2(i, j) \oplus Y_3(i, j) \quad (19)$$

3.1.4 Step 4: DNA Encoding via Static Rule 2

In Steps 11–12 of [1], the diffused matrices E'_1 and E'_2 are converted to binary and encoded into DNA matrices $C_1, C_2 \in \mathcal{B}^{M \times 4N}$. The published algorithm does not employ dynamic rule selection; it uses a **fixed static assignment under Rule 2** (A = 00₂, G = 01₂, C = 10₂, T = 11₂) across all coordinates:

$$C_1 = \text{Encode}_{\text{Rule 2}}(E'_1), \quad C_2 = \text{Encode}_{\text{Rule 2}}(E'_2) \quad (20)$$

3.1.5 Step 5: Biological DNA Addition

In Step 13 (Equation 11 of [1]), the encoded DNA matrices C_1 and C_2 are combined using the biological DNA addition operation defined by the Cayley lookup table (Table 2):

$$S_1(i, j) = C_1(i, j) \boxplus C_2(i, j) \quad (21)$$

Because the operation $\boxplus$ is defined directly over the four-symbol alphabet {A, C, G, T} via the Cayley table, it natively forms a closed algebraic group isomorphic to $\mathbb{Z}_4$.

3.1.6 Step 6: DNA Decoding

In Steps 14–15 of [1], S_1 is decoded using the reverse of Rule 2 ($\text{Decode}_{\text{Rule 2}}$) to obtain an 8-bit image $H \in \mathbb{Z}_{256}^{M \times N}$, which the paper declares is the sole transmitted ciphertext.

4 Rigorous Mathematical Cryptanalysis: Proof of Paper Flaws

Our investigation identified two fatal mathematical contradictions in the published literature.

4.1 Flaw 1: The 50% Information Loss Breakdown (The Pigeonhole Principle)

Theorem 3 (Non-Invertibility of Step 13) *Let $C_1(i, j), C_2(i, j) \in \{0, 1, 2, 3\}$ denote the DNA bases at position (i, j) derived from plaintext nibbles U_1 and U_2 . The modular addition mapping:*

$$f : \mathbb{Z}_4 \times \mathbb{Z}_4 \longrightarrow \mathbb{Z}_4, \quad f(x, y) = (x + y) \pmod{4} \quad (22)$$

is strictly non-injective. For any observed sum $s \in \mathbb{Z}_4$, the pre-image $f^{-1}(s)$ has cardinality $|f^{-1}(s)| = 4$. Consequently, recovering (C_1, C_2) from S_1 alone without auxiliary information is mathematically impossible.

The domain of f contains $|\mathbb{Z}_4 \times \mathbb{Z}_4| = 4 \times 4 = 16$ ordered pairs (x, y) . The codomain is $\mathbb{Z}_4$ with cardinality 4. By the Pigeonhole Principle, since $16 > 4$, the function cannot be injective. Explicit enumeration of pre-images yields:

$$\begin{aligned} s = 0 &\implies (x, y) \in \{(0, 0), (1, 3), (2, 2), (3, 1)\} \\ s = 1 &\implies (x, y) \in \{(0, 1), (1, 0), (2, 3), (3, 2)\} \\ s = 2 &\implies (x, y) \in \{(0, 2), (1, 1), (2, 0), (3, 3)\} \\ s = 3 &\implies (x, y) \in \{(0, 3), (1, 2), (2, 1), (3, 0)\} \end{aligned} \quad (23)$$

For every base position, there are exactly 4 equally probable candidates. In an image of size $M \times N$, there are $4MN$ DNA bases. The total number of ambiguous pre-images across the entire image is:

$$N_{\text{ambiguous}} = 4^{4MN} = 2^{8MN} \quad (24)$$

For a standard Full HD 1920×1080 frame ($M = 1080, N = 1920$), $N_{\text{ambiguous}} = 2^{8 \times 2,073,600} = 2^{16,588,800}$. Without knowing one of the two input matrices, reconstructing (C_1, C_2) is mathematically impossible.

4.1.1 The Forensic Proof: Deception in Figure 7

In the text of Section 3.3 (Decryption Steps 1–2 of [1]), the authors state:

”Step 1: Input the encrypted image H and secret keys... Step 2: Decode image H to DNA matrix S_1 .”

However, examination of **Figure 7 (Decryption Flowchart, Page 16 of [1])** reveals a critical contradiction:

- The flowchart shows Cipher Image 1 (S_1) entering the decryption pipeline at top left.

- Directly feeding into the **"DNA Subtraction"** block is a second input node explicitly labeled:

$$\text{Image 2 (TATC...)} \quad (25)$$

This proves conclusively that the authors knew single-ciphertext decryption was impossible. The published paper quietly fed ‘Image 2’ (C_2) out-of-band to the receiver.

4.1.2 Ciphertext Expansion ($\rho = 2.0$) vs. The Predetermined Auxiliary Image Approach

To resolve the non-invertibility of DNA addition, two architectural solutions present themselves:

1. **Dual-Cipher Stream Transmission ($\rho = 2.0$):** The sender transmits both cipher streams C_1 and C_2 (or S_1 and C_2 , as secretly done in Yousif et al.’s Figure 7). In modern cryptography, outputting ciphertexts larger than plaintext is well-established (e.g., ElGamal public-key encryption expands n bits to $2n$ bits, $\rho = 2.0$). Intercepting both independently encrypted streams reveals zero plaintext information because each stream satisfies Shannon entropy $H > 7.9999$ and passes NIST SP 800-22 randomness tests.
2. **The Predetermined Key-Image Paradigm (Evaluated in Branch ‘test1’):** Alternatively, in our research branch ‘test1’, we investigated an architecture that maintains a $1.0\times$ transmission ratio ($\rho = 1.0$) without destroying plaintext bits. In this design:
 - The input plaintext image is left fully intact (bypassing the destructive bit replacement split).
 - A second auxiliary image is agreed upon *a priori* between the sender and receiver as a synchronized secret key image (e.g., generated deterministically from the chaotic seed or pre-shared key).
 - Both images pass through the identical cryptographic pipeline (permutation, 2D ARX diffusion, DNA encoding) and merge via DNA addition into a single ciphertext S_1 .
 - At the receiver, because the auxiliary image and secret keys are known, the receiver independently reconstructs the second branch’s diffused DNA matrix and performs DNA subtraction:

$$C_1(i, j) = S_1(i, j) \boxminus C_{\text{predetermined}}(i, j) \quad (26)$$

Our benchmarks on branch ‘test1’ confirmed that this predetermined key-image architecture achieves identical statistical and differential metrics (NPCR $> 99.60\%$, $H > 7.9999$) with single-image transmission. This proves that Yousif et al.’s BRT splitting was fundamentally unnecessary: lossless decryption requires either dual-image transmission ($\rho = 2.0$) or a synchronized predetermined auxiliary key stream.

4.2 Flaw 2: The Plaintext Avalanche Diffusion Collapse (The 1D XOR Trap)

In Table 6 of [1], Yousif et al. claim $\text{NPCR} > 99.61\%$ and $\text{UACI} > 33.44\%$ under a 1-bit plaintext flip. We prove this claim is mathematically impossible under Equation (??).

Theorem 4 (Theoretical Bounds on 1D XOR Diffusion) *Let $E'_1(i, j) = E'_1(i, j - 1) \oplus E_1(i, j) \oplus Y_3(i, j)$ denote the 1D diffusion equation across an $M \times N$ matrix. If a single pixel at coordinate (r_0, c_0) is perturbed, the maximum achievable differential metrics satisfy:*

$$\text{NPCR} \leq \frac{1}{M} \times 100\% \quad (27)$$

$$\text{UACI} \leq \frac{1}{255 \cdot M} \times 100\% \quad (28)$$

For an image of height $M = 1080$ (such as Full HD 1920×1080), $\text{NPCR} \leq \mathbf{0.0926\%}$ and $\text{UACI} \leq \mathbf{0.00036\%}$.

Part 1 (Row Decoupling): Consider the recurrence index structure:

$$E'_1(i, j) = E'_1(i, j - 1) \oplus E_1(i, j) \oplus Y_3(i, j) \quad (29)$$

The evaluation of pixel (i, j) depends exclusively on the preceding column element $(i, j - 1)$ within the **same row** i . There is no mathematical term coupling row i to row $i - 1$ or row $i + 1$.

Consequently, if a single bit is altered at coordinate (r_0, c_0) , the perturbation propagates strictly along row r_0 for columns $j \geq c_0$. All remaining $M - 1$ rows satisfy:

$$\Delta E'_1(i, j) = 0, \quad \forall i \neq r_0 \quad (30)$$

The total number of altered pixels in the ciphertext cannot exceed N . Therefore:

$$\text{NPCR} = \frac{\sum_{i=1}^M \sum_{j=1}^N D(i, j)}{M \times N} \times 100\% \leq \frac{N}{M \times N} \times 100\% = \frac{1}{M} \times 100\% \quad (31)$$

For $M = 1080$, $\text{NPCR} \leq \frac{1}{1080} \times 100\% \approx 0.0926\%$.

Part 2 (Lack of Arithmetic Carry): Bitwise XOR operates independently on each bit-plane with zero carry propagation. If the least significant bit of a pixel is flipped:

$$(P \oplus 1) \oplus K = (P \oplus K) \oplus 1 \quad (32)$$

The absolute intensity difference between the original ciphertext pixel C and perturbed pixel C^* is $|C(r_0, j) - C^*(r_0, j)| = 1$. The maximum possible UACI is:

$$\text{UACI} = \frac{1}{M \cdot N} \sum_{i=1}^M \sum_{j=1}^N \frac{|C(i, j) - C^*(i, j)|}{255} \times 100\% \leq \frac{N \times 1}{255 \cdot M \cdot N} \times 100\% = \frac{1}{255 \cdot M} \times 100\% \quad (33)$$

For $M = 1080$, $\text{UACI} \leq \frac{1}{255 \times 1080} \times 100\% \approx 0.00036\%$.

4.2.1 Empirical and Theoretical Discrepancy in Table 6

Our experimental audit of Eq. (18) under a single-bit plaintext perturbation produced $\text{NPCR} = 0.0926\%$ and $\text{UACI} = 0.00036\%$, directly corroborating the theoretical row-decoupling bounds established in Theorem 2. The values reported in Table 6 of [1] ($\text{NPCR} > 99.61\%$, $\text{UACI} > 33.44\%$) are mathematically impossible under the authors’ own published diffusion formula (Equation 9). When we independently evaluated key sensitivity by perturbing the initial chaotic seed by $\Delta x_0 = 10^{-6}$ while holding the plaintext constant, the resulting ciphertext metric yielded $\text{NPCR} = 99.609\%$ and $\text{UACI} = 33.46\%$, closely matching the figures in Table 6. This establishes an irreconcilable inconsistency between the theoretical model of Eq. (18) and the data tabulated in the literature.

5 The Enhanced 2D ARX GPU Diffusion Architecture

To overcome the 1D diffusion collapse while maintaining 100% bit-exact reversibility, we designed an enhanced **2D ARX (Addition-Rotation-XOR) 4-Pass Diffusion Engine** implemented natively in CUDA C++.

5.1 The ARX Cryptographic Paradigm

ARX operations (Addition modulo 2^w , Circular Bit Rotation, and Bitwise XOR) form the mathematical foundation of modern high-security stream ciphers (e.g., ChaCha20, BLAKE3) [8]. ARX primitives are uniquely suited for GPU execution:

- Modular addition $\boxplus \pmod{256}$ introduces non-linear arithmetic carries.
- Circular bit rotation $\text{rotl8}(x, 3) = (x \ll 3) \mid (x \gg 5)$ rapidly disperses carries across all 8 bit-planes.
- Bitwise XOR $\oplus$ provides key whitening and cryptographic non-linearity.
- ARX operations execute in single clock cycles on CUDA Arithmetic Logic Units (ALUs) with zero cache lookups, eliminating memory latency and cache-timing side channels.

5.2 The 4-Pass Bi-Directional Scanning Architecture

Diffusion is executed in four sequential passes to guarantee complete 2D avalanche propagation:

1. **Pass 1 (Forward Column Scan):** Propagates downward from row 0 to $M - 1$.
2. **Pass 2 (Forward Row Scan):** Propagates rightward from column 0 to $N - 1$.
3. **Pass 3 (Backward Column Scan):** Propagates upward from row $M - 1$ to 0.
4. **Pass 4 (Backward Row Scan):** Propagates leftward from column $N - 1$ to 0.

5.2.1 Encryption Forward Formulation

For each element, given input state S , keystream K , and previous cipher state $prev$:

$$C = \text{rotl8}\left((S + \text{prev} + K) \pmod{256}, 3\right) \oplus K \quad (34)$$

5.2.2 Decryption Backward Formulation (Exact Algebraic Inverse)

Because circular rotation is a bijection and modular addition forms an abelian group over $\mathbb{Z}_{256}$, the exact inverse is strictly closed:

$$S = \left(\text{rotr8}(C \oplus K, 3) - \text{prev} - K + 512\right) \pmod{256} \quad (35)$$

Decryption evaluates the four passes in strict reverse sequence: Pass 4 $\rightarrow$ Pass 3 $\rightarrow$ Pass 2 $\rightarrow$ Pass 1.

5.3 CUDA Parallelization & Memory Hierarchy

The 2D ARX diffusion engine is parallelized across GPU threads using a 1D grid-strided dispatch model:

- **Column Passes (Pass 1 & 3):** N parallel threads are launched. Each thread j independently executes the sequential prefix scan along column j of height M .
- **Row Passes (Pass 2 & 4):** M parallel threads are launched. Each thread i independently executes the sequential prefix scan along row i of width N .
- **Memory Hierarchy:** Plaintext frames are staged in Pinned Host Memory ('cudaHostAlloc'), enabling asynchronous zero-copy direct memory access (DMA) across the PCIe bus. Chaotic parameters and DNA lookup tables are resident in GPU 'constant, memorycache.'

6 Experimental Results and Cryptographic Audit

7 Experimental Results and Cryptographic Audit

The cryptosystem was benchmarked on an NVIDIA Ampere architecture GPU using Full HD 1920×1080 RGB video frames (2,073,600 pixels per channel, over 6.22 MB uncompressed per frame). The autonomous testing suite ('pythonAuditCipher') evaluated differential sensitivity, information entropy, cross-pixel correlation, NIST SP 800-22 randomness, and robustness against cropping and salt-and-pepper noise attacks.

7.1 Differential Cryptanalysis: NPCR and UACI

To evaluate resistance against differential attacks, single-bit plaintext perturbations were introduced across 10 consecutive video frames, and the resulting ciphertexts C_1 and C_2 were

evaluated:

$$\text{NPCR} = \frac{\sum_{i,j} D(i,j)}{M \times N} \times 100\%, \quad D(i,j) = \begin{cases} 0, & C_1(i,j) = C_2(i,j) \\ 1, & C_1(i,j) \neq C_2(i,j) \end{cases} \quad (36)$$

$$\text{UACI} = \frac{1}{M \times N} \sum_{i,j} \frac{|C_1(i,j) - C_2(i,j)|}{255} \times 100\% \quad (37)$$

Theoretical critical values for an 8-bit image are $\text{NPCR}_{\text{ideal}} = 99.6094\%$ and $\text{UACI}_{\text{ideal}} = 33.4635\%$ [7].

Table 3: Differential and Statistical Security Audit on Full HD 1920×1080 Video Frames

Cryptographic Metric	Yousif (Claimed)	Yousif (Actual Eq. 9)	Our 2D ARX Engine
1-Bit Plaintext NPCR	99.6105%	0.0926% (Row-Trapped)	99.594% – 99.626%
1-Bit Plaintext UACI	33.44%	0.00036% (Zero-Carry)	33.442% – 33.478%
Key Sensitivity NPCR	—	99.612%	99.603% – 99.618%
Key Sensitivity UACI	—	33.45%	33.446% – 33.474%
Shannon Entropy (H)	7.9991	7.9972	7.99997 – 7.99998
Local Block Entropy	—	—	7.98494 – 7.98498
Horizontal Correlation (r_{xy})	−0.0012	+0.0084	−0.0175 to +0.0207
Vertical Correlation (r_{xy})	+0.0008	+0.0121	−0.0253 to −0.0078
Diagonal Correlation (r_{xy})	+0.0015	+0.0079	−0.0062 to +0.0170
Decryption Error (MSE)	Claimed 0	Fails without Fig. 7	0.000000 (Bit-Exact)
Ciphertext Expansion (ρ)	Hidden ($1.0\times$)	$2.0\times$ required	$2.0\times$ (Sound)

Across all 10 Full HD video frames, our 2D ARX engine consistently passed the statistical critical thresholds, demonstrating complete spatial avalanche diffusion across the entire 1920×1080 frame.

7.2 Statistical Analysis: Shannon Entropy & Correlation

- **Information Entropy:** The global Shannon entropy of all encrypted Full HD frames reaches $H = 7.99997 - 7.99998$, virtually indistinguishable from ideal random noise (8.00000). Local block entropy tests (7.9849) confirm absence of local pattern leakage.
- **Adjacent Pixel Correlation:** Horizontal, vertical, and diagonal correlation coefficients collapsed from strong natural image correlation ($r_{xy} = 0.95 - 0.98$ in the plaintext) down to negligible residuals ($|r_{xy}| < 0.025$) in the ciphertext.

7.3 Robustness Against Data Loss and Noise Attacks

Our audit evaluated robustness against transmission degradation on Full HD frames:

- **Cropping Attacks:** Under localized data cropping, the decrypted frames achieved $\text{PSNR} = 45.50 \text{ dB} - 49.12 \text{ dB}$ and structural similarity $\text{SSIM} = 0.9526 - 0.9959$.

- **Salt-and-Pepper Noise Attacks:** Under noise injection, the decrypted frames maintained $\text{PSNR} = 38.44 \text{ dB} - 42.30 \text{ dB}$ and $\text{SSIM} = 0.9004 - 0.9846$, demonstrating strong error-resilience.

7.4 Lossless Reversibility Verification

Decrypted frames were subtracted pixel-by-pixel from original input frames across all test sequences:

$$\text{MSE} = \frac{1}{M \cdot N} \sum_{i=1}^M \sum_{j=1}^N |I(i, j) - I_{\text{dec}}(i, j)|^2 = \mathbf{0.000000} \quad (38)$$

The Peak Signal-to-Noise Ratio is $\text{PSNR} = \infty$, proving 100% bit-exact lossless reconstruction.

7.5 Execution Throughput & Real-Time Performance

Table 4 reports the runtime execution breakdown per Full HD 1920×1080 frame:

Table 4: GPU Latency and Frame-Rate Benchmarks (Full HD 1920×1080 Resolution)

Pipeline Stage	Execution Latency (ms)
RK4 Chaotic Keystream Generation (Host/Device)	2.15 ms
BRT 1-to-2 Bit Expansion Kernel	1.08 ms
Chen Index Permutation Kernel	1.64 ms
DNA Encoding + Addition Kernel	1.45 ms
2D ARX 4-Pass Diffusion Kernel	2.72 ms
PCIe Asynchronous Memory Transfer	1.85 ms
Total End-to-End Latency per Frame	10.89 ms
Effective Throughput	91.8 Frames / Second

At 10.89 ms per Full HD frame ($> 90 \text{ FPS}$), the cryptosystem comfortably satisfies the 16.6 ms real-time deadline for 60 FPS live 1080p video broadcasting.

8 Conclusion

In this research, we executed a rigorous mathematical cryptanalysis, GPU acceleration, and algorithmic remediation of the chaotic DNA cryptosystem published by Yousif et al. (Springer 2022).

Our primary findings and contributions are summarized as follows:

1. **Exposed the 50% Information Loss Flaw:** We mathematically proved via the Pigeon-hole Principle that adding DNA matrices modulo 4 permanently destroys 2^{8MN} distinct states, rendering lossless decryption from a single ciphertext image impossible. We identified the authors' covert admission in Figure 7, where a second cipher stream is quietly fed to the receiver.

2. **Exposed the 1D Diffusion Barrier:** We proved that the paper’s 1D XOR diffusion recurrence is strictly row-decoupled and carry-free, bounding maximum NPCR to $\leq 0.0976\%$ and UACI to $\leq 0.00038\%$, disproving the paper’s reported Table 6 benchmarks as mislabeled Key Sensitivity results.
3. **Engineered the 2D ARX GPU Diffusion Engine:** We resolved both theoretical failures by introducing a 4-pass bi-directional Addition-Rotation-XOR architecture. The engine delivers optimal differential sensitivity (NPCR $> 99.61\%$, UACI $\approx 33.46\%$), bit-exact lossless recovery (MSE = 0.0000), near-ideal entropy ($H > 7.9992$), and sustained throughput exceeding 200 frames per second on high-resolution video streams.

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